

To create a new rule, follow the steps shown below:

- Select where the new rule is to be placed in the chain.
- Right-click the rule in the chain where the new rule is to be located.
- Select either *Insert New Rule* or *Add New Rule Below* (depending upon where the new rule needs to live in the chain).

The new rule will be placed within the chain – it will be fairly generic and will deny all traffic. Obviously, this new rule must be edited.

4. **Compile rules:** You can convert rules into a configuration file for the firewall. After you have created a firewall object and updated the rules in its *Policy* object, the next step is to compile and install these rules on your firewall.

The process is very simple. After you finish building the firewall, perform the following steps:

1. Save the firewall by clicking the *Save* button.
2. Compile the firewall by clicking the *Compile* button and walking through the easy-to-use *Compile Wizard*.
3. Install the firewall by clicking the *Install* button.

As soon as the installation is complete, the firewall will run. This installation will also ensure the firewall runs upon rebooting the machine. If changes are made in the currently running firewall, it is necessary to re-compile and re-install the firewall.

5. **Deploy configuration:** Install the configuration file on the firewall. Deployment is done in two steps.
 - **Compile.** Compiling the rules converts the rules from Firewall Builder's syntax to a file containing the command syntax used by the firewall device type and saves the output in a file. For example, compiling the rules for a firewall with the type set to iptables will generate a file that includes the rules in iptables format. After the firewall rules are compiled you can view the resulting command file generated by Firewall Builder.
 - **Install.** Installing the firewall rules involves transmitting the file with the rules to the target device and then running that file to install the rules. The secure protocols SSH and SCP are used for this, so you will need to provide valid user credentials with the appropriate permissions on the target device.

Features

The following are the features of Firewall Builder:

- Over 100 predefined objects for the most popular and widely used protocols and services, and the capability to create custom objects describing IP, ICMP, TCP, UDP or custom services. Firewall Builder can also create objects to describe hosts, networks and address ranges.
- Powerful wizards and other tools to help you jumpstart your firewall policy, and do it right from the first try. Wizards implement standard policies for typical network layouts, which can later be extended and edited by hand.



Figure 2: Firewall Builder working

- The network discovery tool which automates the creation of multiple objects.
- An object-oriented approach to policy management: Any change made to an object gets immediately reflected in all the policy rules of all firewalls using that object. The configuration for multiple firewalls shares the same object database.
- Generates scripts that set up interfaces, IP addresses, snmp, ntp and logging parameters and other aspects of configuring the firewall machine.
- Convenient GUI that supports standard copy/paste operations for both text strings and network objects, and allows for policy editing using the drag-and-drop feature.
- Configuring versioning control using RCS.
- Designed to manage both dedicated remote firewalls and local firewall configurations for servers, workstations and laptops.
- Supports several firewall platforms, including Cisco PIX and open source firewalls like iptables, ipfilter and pf. It has the ability to print a single object, a firewall policy, the whole hierarchy, or export it to the text file in either plain text or HTML format.

Firewall Builder hides a lot of complexity but it also hides a part of iptables' power within itself. It is an incredibly powerful and flexible security tool that any Linux administrator should get to know. This tool is far better at creating firewalls as compared to GUI tools like Gufw and other CUI based tools like iptables. **END** 🐧

References

- [1] <http://fwbuilder.sourceforge.net/>
- [2] <https://github.com/fwbuilder>

By: Anand Nayyar

The author works at Duy Tan University in Vietnam. He loves to explore open source technologies, sensor communications, network security and the Internet of Things. He can be reached at anandnayyar@duytan.edu.vn. YouTube channel: *Gyaan with Anand Nayyar* at www.youtube.com/anandnayyar.